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# *International Mathematics Assessments for Schools (2024-2025)*

## Junior Division Round 2

Time: 120 minutes

*Printed Name* _____ *Code* _____ *Score* _____

## Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

**The following area is to be filled in by the judges;**
**the contestants are not supposed to mark anything here.**

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | Total Score | Signature |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Score | | | | | | | | | | | | | | | | | |
| Score | | | | | | | | | | | | | | | | | |

# Junior Division Round 2

## Questions 1 to 5, 4 marks each

1. A real-valued function $f$ is defined on all real numbers such that $f(x) \le x$ for all real numbers $x$ and $f(x+y) \le f(x) + f(y)$ for all real numbers $x$ and $y$. What is the numerical value of $f(20) + f(25)$?

（A）$-45$　（B）$-2$　（C）0　（D）2　（E）45

Answer：_____

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2. Alex and Boris drive at different constant speeds. At a certain moment in the morning of one day, Alex started to drive from town A to town B along a road, while Boris started to drive from town B to town A along the same road. After they met at 11:00 AM, Alex continued to drive and arrived at town B at 3:00 PM while Boris continued to drive and arrived at town A at 8:00 PM. At what time did they start to drive on that day?

（A）4:30 AM （B）5:00 AM （C）5:30 AM （D）6:00 AM （E）6:30 AM

Answer：_____

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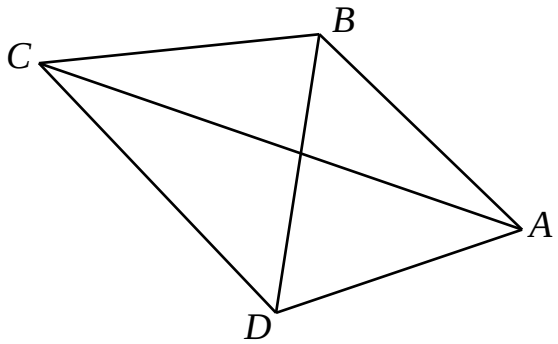
3. There exist some perfect squares where its last four digits (from the right) are all equal to a certain digit $a$. For example, $100^2 = 10000$. How many different possible values of digit $a$ are there?

（A）1　（B）2　（C）3　（D）5　（E）6

Answer：_____

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4. As shown in the diagram below, $ABCD$ is a quadrilateral with $AB = BC$, $\angle ABD = 55°$, $\angle CBD = 75°$ and $\angle ADC = 115°$.

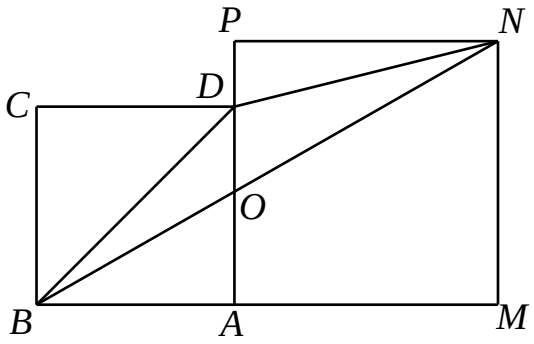



What is the positive difference (subtract the smaller angle from the larger angle), in degrees, of $\angle BAD$ and $\angle BCD$?

（A）0°　（B）10°　（C）11.5°　（D）13°　（E）14.5°

Answer：_____

5. As shown in the diagram below, squares *ABCD* and *APNM* are placed side by side, without overlapping. The side length of square *ABCD* is less than the side length of square *APNM* and *O* is the intersection of *BN* and *AD*.




Only one statement from options (A) - (E) is **ALWAYS** true. Which one is it?

（A）The areas of triangles *BCD* and *DON* are equal.

（B）The area of triangle *BDN* is less than the area of triangle *BCD*.

（C）The area of triangle *BCD* is larger than the area of triangle *DPN*.

（D）The area of triangle *BOA* is less than the area of triangle *DON*.

（E）The areas of triangles *BOA* and *DON* are equal.

Answer：_____

## Questions 6 to 13, 5 marks each

6. A two-digit number is said to be ***good*** if it is equal to the sum of squares of each of its digits minus the product of its digits. For example, 48 is a ***good*** number since $48 = 4^2 + 8^2 - 4 \times 8$. What is the sum of all the ***good*** numbers, including 48?

Answer：_____

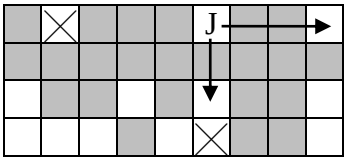
7. A three-digit number is said to be ***interesting*** if it satisfies both the conditions below:
   - All digits of this number are non-zero.
   - After removing any one of the digits and adding the removed digit to the remaining two-digit number, the resulting number is always divisible by 8.

   For example, 195 is an interesting number because if we remove 1, then $1 + 95 = 96$ is divisible by 8, and if we remove 9, then $15 + 9 = 24$ is also divisible by 8, and if we remove 5, then $19 + 5 = 24$ is also divisible by 8.

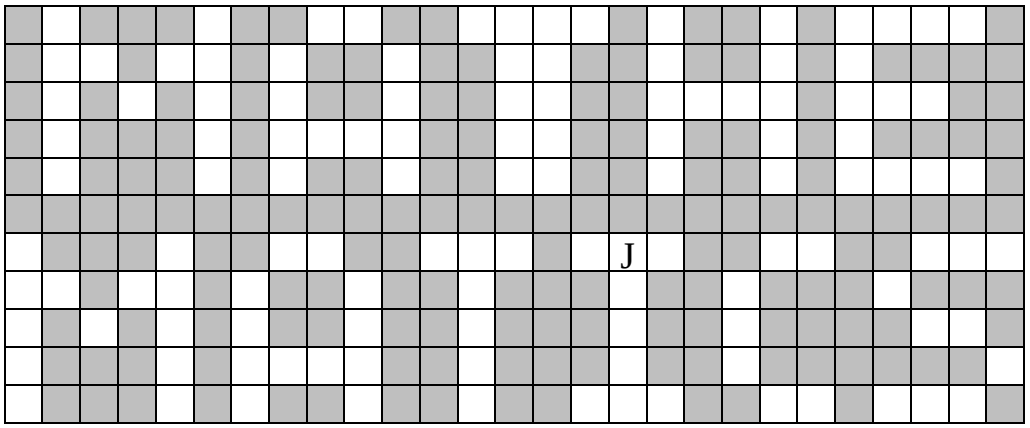
   How many ***interesting*** three-digit numbers are there in total?

Answer：_____

8. In a computer game, a maze is illustrated as a grid, where some cells are painted in grey. The grey cells indicate where traps have been placed and the white cells indicate safe cells. A jumping man named Jay is standing on a white cell, which is indicated as "J". Jay may jump from one white cell to another white cell. However, he may only jump horizontally or vertically at most 3 cells. He can jump over a sequence of grey cells, but he cannot jump over another white cell (and he cannot land on a grey cell).
For example, as shown in the diagram below, if Jay stands on the white cell which is indicated as "J", he may jump three cells to the right or he may jump two cells down, but he cannot jump four cells to the left or three cells down. Also, if he wants to get to any other white cell, he would need to make additional jumps.

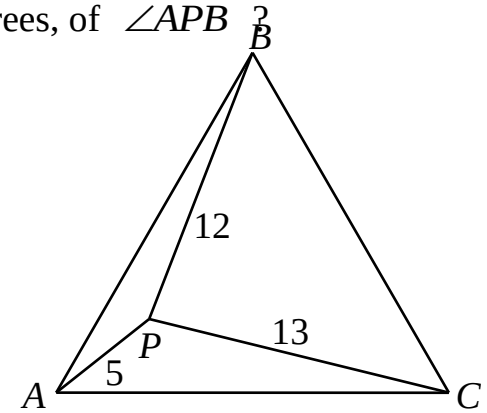



In the following maze shown below, Jay begins on the white cell indicated as "J". What is the minimum number of jumps he needs to do to be sure that any other white cell can be reached when jumping from this starting position?




Answer : _____

9. In equilateral triangle $ABC$, let $P$ be a point inside the triangle such that $PA = 5$ cm, $PB = 12$ cm and $PC = 13$ cm, as shown in the diagram below. What is the measure, in degrees, of $\angle APB$ ?

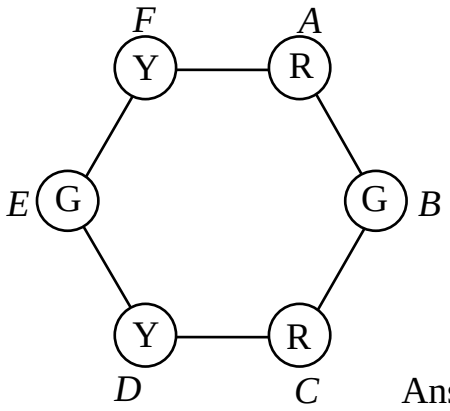



Answer : _____ °

10. There are 18 cards. Each card is labelled with exactly one positive integer from the set {1, 2, 3, 4, 5, 6, 7, 8, 9} so that two cards are labelled 1, another two cards are labelled 2, …, another two cards are labelled 8 and the last two cards are labelled 9. Amy and Brandon play a game by using these cards. Amy randomly selects two cards at the same time. After that, Amy just tells Brandon the value of the product of the numbers on the selected cards. If Brandon can determine the two numbers on the cards selected by Amy in one try without guessing, then Brandon wins the game. What is the probability, $\frac{q}{p}$, where $p$ and $q$ are relatively prime, that Brandon wins the game?

Answer : _____

11. In how many ways can we paint each of the six vertices of hexagon *ABCDEF* using one of the three colours: red, green and yellow, such that no two adjacent vertices have the same colour? (One way of colouring is shown in the diagram below, where R means red, G means green and Y means yellow.)




Answer : _____ ways

12. In a ball pit, there are sufficiently many balls of four colours: red, blue, yellow and green. Now, 2025 persons will each take turns, one at a time, to take out four balls from the ball pit. What is the minimum number of persons that will have exactly the same set of colours and same number of balls of each colour? Provide the tightest bound.

Answer : _____ people

13. In a store, each cup costs \$21, each bowl costs \$24 and each stockpot costs \$168. One day, some cups, bowls and stockpots were sold for a total of \$2025 such that at least one of each of these three products was sold. How many different possible sales states were there for these three products on that day?

Answer : _____ sales states

## Questions 14 to 15, 20 marks each
## (Detailed solutions are needed for these two problems)

14. A recurring decimal is a decimal representation of a number whose digits are periodic in nature, meaning that, after some places, the same sequence of digits repeats continuously. For example, $\frac{1234}{9999} = 0.12341234... = 0.\dot{1}2\dot{3}4\dot{}$. To calculate the sum of all the digits of a recurring decimal, aside from summing all the digits directly, there is another method as following:

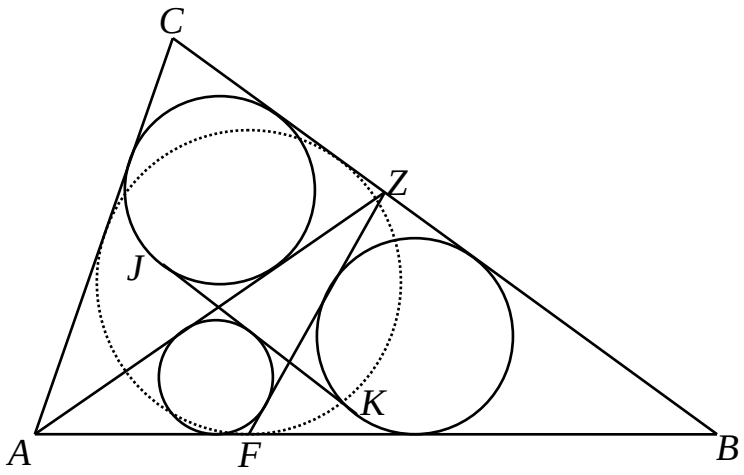
(a) It is known that the recurring decimal of $\frac{1}{17}$ is 16 digits long, which you can assume to be true. To find the sum of all the digits of the recurring decimal of $\frac{1}{17}$, we can suppose that $\frac{1}{17} = \frac{A}{10^{16}-1}$, where positive integer $A$ is the repetend. Since the period is 16, we have $10^{16}-1$ is divisible by 17 but $10^k - 1$ is not divisible by 17 for $1 \le k < 16$. Notice that 17 is a prime number, by Fermat's Little Theorem, we have $10^{16} \equiv 1 \pmod{17}$ . Thus, $10^{\frac{16}{2}} = 10^8 \equiv \pm 1 \pmod{17}$.

Now, write down the subsequent proof to show that the answer is 72. (5 marks)

14. (b) It is known that the recurring decimal of $\frac{1}{179}$ is 178 digits long, which you can assume to be true. What is the sum of all the digits of the recurring decimal of $\frac{1}{179}$? Please use a similar method as part (a). (15 marks)

Answer：_____

15. Let $F$ be the point of tangency of the incircle of the triangle $ABC$ with $AB$, and let $Z$ be an arbitrary point strictly between $B$ and $C$, as shown in the diagram below. Let $K$ and $J$ be the point on the incircles of triangles $BFZ$ and $CAZ$, respectively, such that $KJ$ is the other exterior common tangent with these two circles. Prove that $KJ$ is also a tangent of the incircle of triangle $AFZ$.




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